

Filip Bělík

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PhD Candidate

Department of Mathematics and Scientific Computing and Imaging Institute
University of Utah, Salt Lake City, USA

EDUCATION

• University of Utah

Mathematics PhD: Applied & Computational Mathematics

August 2022 - Present

Salt Lake City, UT, USA

- Co-advised by [Dr. Akil Narayan](#) and [Dr. Christel Hohenegger](#)
- Intended graduation in May 2027
- President of University's [SIAM Student chapter](#), Fall 2024-Spring 2026
- Organized [Wasatch SIAM Student Chapters Conference](#), Spring 2026
- Chair of Mathematics Graduate Student Recruitment Committee, Spring 2025
- GPA: 4.00

• Gustavus Adolphus College

BA Honors Mathematics & BA Computer Science

September 2018 - May 2022

St. Peter, MN, USA

- Student host for [Nobel Conference 2021, Big Data](#)
- Mathematics, Computer Science, and Statistics (MCS) Department Assistant, 2021
- President of Club Tennis; Co-President of Coding Club; MCS Club; Running Club
- Cumulative GPA: 3.989
- Major GPA: 4.00

PUBLICATIONS/PREPRINTS

1. Filip Bělík, Akil Narayan, and John Turnage. *Constructive Tchakaloff results and well-conditioned quadrature through randomized least squares*. 2026. arXiv: [2609.09506 \[math.NA\]](#). URL: <https://arxiv.org/abs/2609.09506>
2. Henry Crandall et al. "Cuffless hemodynamic monitoring with physics-informed machine learning models". en. In: *Nat. Commun.* (May 2026)
3. Filip Bělík, Yanlai Chen, and Akil Narayan. *Greedy Rational Approximation for Frequency-Domain Model Reduction of Parametric LTI Systems*. 2025. arXiv: [2512.23814 \[math.NA\]](#). URL: <https://arxiv.org/abs/2512.23814>
4. Filip Bělík, Jesse Chan, and Akil Narayan. *Efficient and Robust Carathéodory-Steinitz Pruning of Positive Discrete Measures*. 2025. arXiv: [2510.14916 \[math.NA\]](#). URL: <https://arxiv.org/abs/2510.14916>

TEACHING

• Engineering Calculus II

Instructor of Record

Spring 2026

University of Utah

• Engineering Calculus I

Instructor of Record

Spring 2025

University of Utah

• Graduate Applied Linear Operators and Spectral Methods

Teaching Assistant

Fall 2025

University of Utah

• Graduate Numerical Linear Algebra

Teaching Assistant

Spring 2024

University of Utah

• Mathematics of Medicine

Laboratory Instructor

Spring 2023

University of Utah

• Various courses in Mathematics, Computer Science, and Statistics Department

Tutor, Teaching Assistant, and Grader

Spring 2019 - Spring 2022

Gustavus Adolphus College

WORK EXPERIENCE

• University of Utah Mathematics Department

Graduate Research and Teaching Assistant

August 2022 - Present

Salt Lake City, UT, USA

- Instructor of record for MATH 1320, Engineering Calculus II, Spring 2026
- Instructor of record for MATH 1310, Engineering Calculus I, Spring 2025
- Research funding under Dr. Narayan, Dr. Christel Hohenegger, and a University of Utah funding incentive seed grant
- Lab TA for MATH 4600 Mathematics in Medicine

• Allianz Life

Hedging Intern

May 2021 - August 2021 & May 2026 - August 2026

Golden Valley, MN, USA

- Software development with C# and SQL
- Implemented procedure for automating the labeling of incoming market data
- WinApps development for sparse 3D data and assistance with software development life cycle
- Study of mathematical finance and software implementation for delta and gamma hedging

• Gustavus Mathematics and Computer Science Department

TA & Tutor & Grader

February 2019 - May 2022

St. Peter, MN, USA

- Computer Science I Teacher's Assistant, Grader, and Tutor (Python)
- Computer Science II Teacher's Assistant and Tutoring (Kotlin and Java)
- Discrete Mathematics Grader
- Online volunteer tutoring during COVID semester

• Her Next Play

CRM Intern

June 2020 - August 2020

Remote

- Research and implementation of contact resource management software
- Presentation and instruction to executives

CONFERENCES

• Wasatch SIAM Student Chapters Conference

University of Utah, Salt Lake City UT

April 2026



- Talk: Frequency Domain Greedy Model Order Reduction for LTI Systems

• SIAM Uncertainty Quantification

Hyatt Regency, Minneapolis MN

March 2026

• Joint Mathematics Meetings

Walter E. Washington Convention Center, Washington D.C.

January 2026



- Talk: Greedy Rational Approximation for Model Reduction of LTI Systems

• Wasatch SIAM Student Chapters Conference

Utah State University, Logan UT

April 2025



- Talk: Carathéodory-Steinitz Pruning for Numerical Quadrature

• MAA North Central Section Meeting

St. Olaf College, Northfield MN

March 2025



- Talk: Carathéodory-Steinitz Pruning for Numerical Quadrature

• Model Reduction and Surrogate Modeling

Scripps Seaside Forum, La Jolla CA

September 2024



- Talk: Greedy Frequency Domain Model Reduction for Parametric Systems: New Theory and Algorithms

• NSF Computational Mathematics PI Meeting

University of Washington, Seattle WA

July 2024



- Poster: Dynamic Bulk Conductivity in Radial Artery

• Mathematical Opportunities in Digital Twins

George Mason University, Arlington VA

December 2023



- Poster: Dynamic Bulk Conductivity in Radial Artery

• Midstates Consortium for Math and Science

University of Chicago, Chicago IL

November 2019



- Talk: The Propagation of Health-Related Habits on Twitter

HONORS AND AWARDS

- **Outstanding Graduate Student** May 2025
University of Utah Mathematics Department
- **RTG: Optimization and Inversion Summer Grant** May 2023
University of Utah Mathematics Department 
- **Fulbright Canada Mitacs Globalink Research Internship** April 2021
Mitacs Globalink 
- **Hilding Research Fund** April 2020
Gustavus Adolphus College
- **First-Year Research Experience (FYRE)** April 2019
Gustavus Adolphus College 
- **Math Problem Solving Competition** 2018, 2019
Gustavus Adolphus College 
- **Dean's Scholarship** Fall 2018
Gustavus Adolphus College

SKILLS

- **Programming Languages:** Julia, Python, MATLAB, LaTeX, C, C++, Java, C#
- **Computer Operating Systems:** Windows, Linux, Mac
- **Web Development:** HTML, JavaScript, CSS
- **Mathematical/Technical Tools:** Jupyter, LaTeX, Maple, SQL
- **Android App Development:** Kotlin
- **Competitive Programming:** ICPC, COMAP, Kattis (fbelik), Project Euler (fbelik)